Scale
~ by weight
6-8 hours
6-8 hours
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Al.4 FUEL/WASTE SPECIFICATION FOR TYPE APPROVAL TEST (~ BY WEIGHT)
Sludge oil consisting of:
Solid waste (class 2)
consisting of:
Classes of waste
75% sludge oil from heavy fuel oil
5% waste lubricating oil
20% emulsified water
50% Food Waste
50% rubbish Containing
Approx. 30% paper,
" 40~ Cardboard,
"
"
10~ Rags,
20~ Plastic
The mixture will have up to 50%
moisture and 7% incombustible
solids
Reference: Waste Classification from Incinerator Institute of America.

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SOURCE: MEPC.59(33).pdf
50% rubbish Containing
Approx. 30% paper,
" 40~ Cardboard,
"
"
10~ Rags,
20~ Plastic
The mixture will have up to 50%
moisture and 7% incombustible
solids
Reference: Waste Classification from Incinerator Institute of America.
Class 0 Trash, a mixture of highly combustible waste such as paper,
cardboard, wood boxes, and combustible floor sweepings, with up to
10% by weight of plastic bags, coated paper, laminated paper,
treated corrugated cardboard, oil rags, and plastic or rubber
scraps. This type of waste contains up to 10% moisture, 5%
incombustible solids and has a heating value of about 19,700 kJ/kg
as fired.
Class 1 Rubbish, a mixture of combustible waste such as paper, cardboard
cartons, wood scrap, foliage, and combustible floor sweepings. The

mixture contains up to 20% by weight of galley or cafeteria waste,
but contains little or no treated papers, plastic, or rubber
wastes. This type of waste contains 25% moisture, 10% incombustible
solids and has a heating value of about 15,100 kJ/kg as fired.
Class 2 Refuse, consisting of approximately even mixture of rubbish and
garbage by weight. This type waste is common to passenger ships

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SOURCE: MEPC.59(33).pdf
solids and has a heating value of about 15,100 kJ/kg as fired.
Class 2 Refuse, consisting of approximately even mixture of rubbish and
garbage by weight. This type waste is common to passenger ships
occupancy, consisting of up to 50% moisture, 7% incombustible solids
and has a heating value of about 10,000 kJ/kg as fired.
Class 3 Garbage, consisting of animal and vegetable wastes from restaurants,
cafeterias, galleys, sick bays and like installations. This type of
waste contains up to 70% moisture, up to 5% incombustible solids and
has a heating value range of about 2,300 kJ/kg as fired.
Class 4 Aquatic life forms and animal remains, consisting of carcasses,
organs and solid organic wastes from vessels carrying animal type
cargos, consisting of up to 85% moisture, 5% incombustible solids
and having a heating value range of about 2,300 kJ/kg as fired.
Class 5 By-product waste, liquid or semi-liquid, such as tar, paints,
solvents, sludge, oil, waste oil, etc., from shipboard operations.
BTU values must be determined by the individual materials to be
destroyed.
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SOURCE: MEPC.59(33).pdf
solvents, sludge, oil, waste oil, etc., from shipboard operations.
BTU values must be determined by the individual materials to be
destroyed.
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Class 6 Solid by-product waste, such as rubber, plastics, wood waste, etc.,
from industrial operations. BTU values must be determined by the
individual materials to be destroyed.
Calorific values kJ/Kg kcal/kg
Vegetable and putrescibles 5,700 1,360
Paper 14,300 3,415
Rag 15,500 3,700
Plastics 36,000 8,600
Oil sludge 36,000 8,600
Sewage sludge 3,000 716

Densities kg/m 3
Paper (loose) 50
Refuse (75% wet) 720
Dry rubbish 110
Scrap wood 190
Wood sawdust 220
Density of loose general waste generated on board ship will be about
130 kg/m 3
Al.5 REQUIRED EMISSION STANDARDS TO BE VERIFIED BY TYPE APPROVAL TEST
02 in combustion chamber 6 - 12%
CO in flue gas maximum average 200 mg/MJ
Soot number maximum average BACHARACH 3 or RINGELMAN 1
Unburned components in
ash residues
Combustion chamber flue gas
outlet temperature range

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SOURCE: MEPC.59(33).pdf
02 in combustion chamber 6 - 12%
CO in flue gas maximum average 200 mg/MJ
Soot number maximum average BACHARACH 3 or RINGELMAN 1
Unburned components in
ash residues
Combustion chamber flue gas
outlet temperature range
(A higher soot number is acceptable
only during very short periods such
as starting up)
Max 10% by Weight
900 - 1,200°C
A high temperature in the actual combustion chamber/zone is an absolute requirement in order to obtain a complete and smoke free incineration, including that of plastic and other synthetic materials while minimizing DIOXINE, VOC (Volatile Organic Compounds), and emissions.
Al.6 FUEL RELATED EMISSION
Al.6.1 Even with good incineration technology the emission from an incinerator will depend on the type of material being incinerated. If for instance a vessel has bunkered a fuel with high sulphur content, then sludge oil from separators which is burned in the incinerator will lead to emission of Sox. But again, the Sox emission from the incinerator would only amount to less than one per cent of the Sox discharged with the exhaust from main and auxiliary engines.
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SOURCE: MEPC.59(33).pdf
of Sox. But again, the Sox emission from the incinerator would only amount to less than one per cent of the Sox discharged with the exhaust from main and auxiliary engines.

A1.6.2 Principal organic constituents (POC) cannot be measured on a continuous basis. Specifically, there are no instruments with provision for continuous time telemetry that measures POC, HCl, or waste destruction efficiency, to gate. These measuremepts can only be made using grab sample
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approaches where the sample is returned to a laboratory for analys~s. In the case of organic constituents (undestroyed wastes), the laboratory work requires considerable time to complete. Thus, continuous emission control can only be assured by secondary measurements.
A1.6.3 ON-BOARD OPERATION/EMISSION CONTROL
For a shipboard incinerator with IMO TYPE APPROVAL, emission control/monitoring should be limited to the following:
.1 Control/monitor 02 content in combustion chamber (spot checks only) .

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